Lightweight Res-CNN-Based Online ESR Estimation Method for Boost Converters

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***Abstract*—This article proposes a lightweight residual learning-based convolutional neural network (Res-CNN) for online real-time estimation of the equivalent series resistance (ESR) of electrolytic capacitors (ECs) in DC/DC boost converters, implemented on an onboard microcontroller unit (MCU) platform. The developed lightweight Res-CNN, which estimates ESR values based on the regression network, shows strong generalization capability under variations in operating and load conditions, together with lightweight model design. A ghost module-based design presented in this article reduces model complexity, including parameter size, model size, and floating-point operations (FLOPs) while preserving diagnostic accuracy. The main contributions can be summarized as follows: 1) regression-based ESR estimation algorithm with strong generalization performance even under untrained device and operating/load conditions; 2) lightweight model architecture designed with a ghost module, which reduces the model complexity; and 3) deployment of the proposed lightweight Res-CNN on an MCU-based online platform. In this article, online real-time ESR estimation is experimentally performed during boost converter operation, confirming consistency between the offline and online evaluation results. The effectiveness of the proposed method is verified with comparative results of previous methods, from the standpoints of estimation accuracy and model complexity.**

***Index Terms*—Boost converter, convolutional neural network (CNN), electrolytic capacitor (EC), equivalent series resistance (ESR) estimation, lightweight model.**

NOMENCLATURE

1D CNN	One-dimensional convolutional neural network.
ADC	Analog to digital conversion.
CNN	Convolutional neural network.
DUT	Device under test.
EC	Electrolytic capacitor.
ESR	Equivalent series resistance.
FLOPs	Floating-point operations.
GAP	Global average pooling.
GPR	Gaussian process regression.
GPU	Graphics processing unit.
KNN	*k*-nearest neighbor.
LSTM	Long short-term memory.
MCU	Microcontroller unit.
MOSFET	Metal oxide semiconductor field effect transistor.
PC	Personal computer.
PV	Photovoltaic.
PWM	Pulsewidth modulation.
ReLU	Rectified linear unit.
Res-CNN	Residual learning-based CNN.
RRMSE	Relative root-mean-square error.
SVM	Support vector machine.
TIHCL	Temporal-level and instance-level hierarchical contrastive learning.
XCCN	Extended convolutional capsule network.

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I. INTRODUCTION

DC/DC converters are power electronic converters commonly used in renewable energy, aerospace, wind turbines, and electric vehicle applications [1], [2], [3], [4]. In reliability-critical systems, such as electric vehicles and aerospace applications, diagnosis of the state of power conversion components is necessary before faults occur. A capacitor is a critical component that suppresses voltage ripple, absorbs harmonics, and supports power balancing [5], [6]. ECs are widely used because of their capacitance density, cost-effectiveness, and high-voltage range [7], [8]. However, ECs age over time owing to chemical changes and thermal stress [9]. In general, the aging process of ECs involves electrolyte vaporization, leading to parameter drifts from their fresh state. End-of-life of ECs can be generally characterized as a doubling of ESR or a 20 % reduction in capacitance (C) compared to the initial value [10], [11]. In particular, ESR is a strong indicator of EC performance degradation because it is less sensitive to temperature and the variations in ESR during the lifetime of an EC are significant [12]; therefore, techniques for monitoring the state have been investigated by estimating the ESR of ECs.

In general, capacitor condition monitoring is categorized into data-driven and model-based methods. In model-based methods, a mathematical representation of the physical system is constructed and used to diagnose capacitor health [12], [13], [14], [15]. Analytical models for monitoring the states of capacitors in DC/DC converters have been established [12], [13], [14]. Ahmad et al. [15] derive an ESR estimation formula for monitoring the state of a capacitor in a PV-based DC system. Accurate modeling is required in these methods because ESR estimation accuracy depends on the mathematical model.

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TABLE I
SUMMARY OF EXISTING DATA-DRIVEN ESTIMATION METHODS FOR CAPACITOR ESR VALUES

	[17]	[18]	[19]	[20]	[21]	[22]	[23]
Topology	Buck converter					Super-buck converter	Buck converter / Super-buck converter
Test device	Trained device						
Test condition	Trained conditions						Untrained conditions
Accuracy	98.34%	87.5%	94.92%	98.33%	97.67%	99.01%	96.33%
Model structure	TIHCL	KNN	ResNet-50	SVM	LSTM-FCN	LSTM	XCCN
Regression or classification	Classification	Classification	Classification	Classification	Classification	Regression	Classification
Online test	X	X	X	X	X	X	O (JETSON Nano)

Data-driven methods can be regarded as black box approaches [16] that establish an input–output mapping through learning without an analytical model and have been applied to monitor the states of capacitors. Representative studies employing datadriven approaches are listed in Table I. Miao et al. [17], [18] use the output voltage ripple of a buck converter to classify the state of a capacitor into discrete ranges, adopting TIHCL and KNN models, respectively. Han et al. [19] use ResNet-50 to classify the capacitor health states in a buck converter. Liu et al. [20] use SVM to classify the normal and degraded states of the capacitor in a buck converter. Miao et al. [21] propose a degradation detection method for a buck converter using an LSTM fully convolutional network (LSTM-FCN). Wu et al. [22] propose an LSTM-based regression approach for capacitor health monitoring in a superbuck converter, with an evaluation conducted via simulations. In the study by Wang et al. [23], wavelet transforms are applied, and XCCN is used to classify the capacitor health into ranges for both buck and superbuck converters.

In practical applications, the capacitor health is diagnosed on capacitor devices and operating/load conditions that are unseen during training. However, most previous studies test the same capacitor devices and conditions used for training [17], [18], [19], [20], [21], [22]. Although Wang et al. [23] investigate the influence of unseen load conditions for trained capacitor devices, their classification setup employs coarse bins. The gap between the fresh-ESR class and the next class is approximately 130 % of the fresh value, indicating a relatively wide class spacing.

In addition, although online monitoring is crucial for industrial deployment, most prior studies evaluate performance in offline settings [17], [18], [19], [20], [21], [22]. Given the limited computing power and resources available on onboard platforms, compression and optimization of the model are required to satisfy online real-time requirements [24]. In this regard, Wang et al. [23] demonstrate the online feasibility of the graphics processing unit (GPU)-based JETSON Nano. However, this study focuses on a GPU board, and lightweighting strategies for the model are not examined. To address this gap, this study targets an MCU-based environment, employs a lightweight model, and evaluates online real-time ESR estimation performance.

A lightweight Res-CNN is introduced for online real-time ESR estimation of ECs in a DC/DC boost converter. The proposed method is designed to achieve both strong generalization under untrained devices and operating/load conditions and a lightweight structure suitable for onboard implementation. To this end, the model uses the input current and output voltage as the primary inputs while also incorporating the duty ratio and output current to account for the operating and load conditions. Residual learning is applied to improve model stability, and the ghost module is adopted to reduce the parameter size, model size, and floating-point operations (FLOPs). Furthermore, the proposed lightweight Res-CNN is deployed on an MCU-based online platform and experimentally tested through online real-time testing under untrained device and operating/load conditions, similar to those of the PC-based offline platform. Comparison with the offline accuracy demonstrates the proposed method's capability for online real-time ESR estimation. The main contributions of this study are as follows.

1) An ESR estimation algorithm is proposed with strong generalization capability under untrained devices and operating/load conditions.
2) A lightweight network structure is constructed to reduce the model complexity.
3) The proposed method is not limited to offline analysis but is experimentally verified through online real-time deployment on an MCU-based online platform during boost converter operation.

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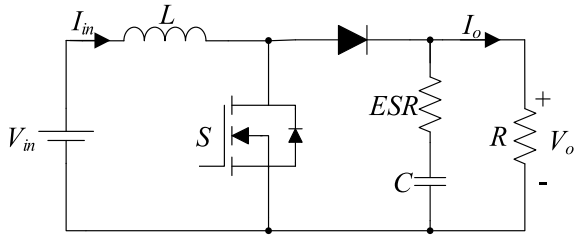


Fig. 1. Topology of the DC/DC boost converter.

To validate the performance of the proposed method, offline experiments are conducted under trained and untrained devices and operating/load conditions, and those results are compared with previous estimation algorithms in terms of accuracy and model complexity. In addition, the online effectiveness of the proposed model, deployed on an MCU platform based on the STMicroelectronics STM32H753ZI, is verified by evaluating the accuracy of online real-time ESR estimation for the capacitor in the boost converter.

II. System Description and Algorithmic Background

A. System Description

The topology considered in this study is that of a boost converter, as shown in Fig. 1. The boost converter operates using PWM, and the output voltage V_o is regulated as a function of the input voltage V_{in} and the duty ratio. In Fig. 1, the input current is I_{in} and the output current is I_o.

B. Algorithmic Background

In general, a 1-D CNN can be effectively applied to process 1-D signals, including time series and sequential data [25], [26]. The output, y_k^l, can be expressed as follows:

$$y_k^l = f\left(\sum_{i=1}^{N_{l-1}} conv1D\left(w_{ik}^{l-1}, s_i^{l-1}\right) + b_k^l\right) \tag{1}$$

where the kernel from the ith neuron of layer $l-1$ to the kth neuron of layer and l is designated as w_{ik}^{l-1}. The output of the ith neuron in layer $l-1$ is denoted by s_i^{l-1}. Furthermore, b_k^l is defined as the bias term for the kth neuron in layer l, while $f(\cdot)$ represents the activation function. Rather than directly learning the mapping $H(x)$ in deep networks, residual learning trains the model to learn the residual $F(x)$ with respect to input x, as expressed by the following equation:

$$F(x) = H(x) - x. \tag{2}$$

Furthermore, the formation of the output y stabilizes the gradient propagation flow, as expressed by [28].

III. Proposed Lightweight Res-CNN

A. Method for Estimating Lightweight Res-CNN-Based ESR

This study proposes a lightweight Res-CNN regression network for capacitor health estimation that integrates residual learning and a ghost module within a 1-D CNN. Fig. 2 shows the overall ESR estimation framework. In this study,

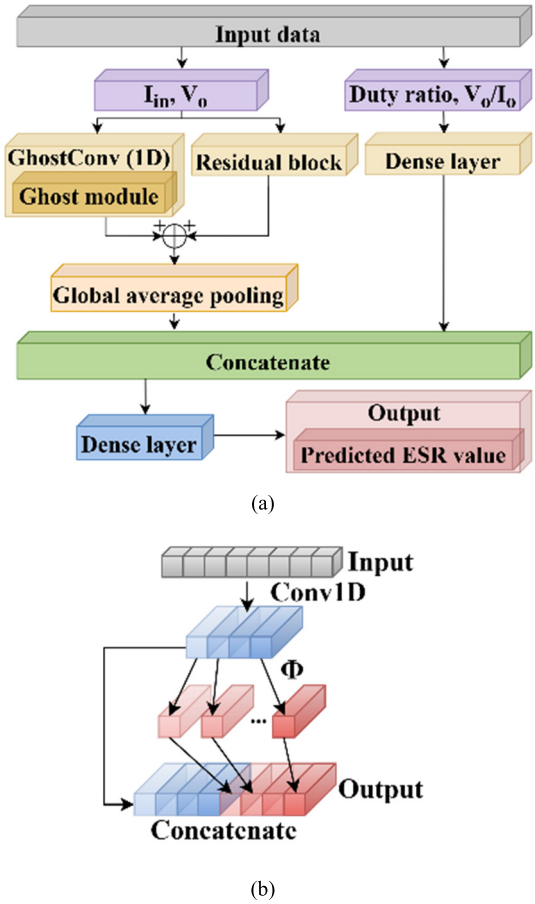


Fig. 2. Proposed ESR estimation method. (a) Architecture of the proposed method. (b) Architecture of a ghost module. Φ represents a cheap operation.

we adopt a regression approach to directly predict the ESR as a continuous value. The detailed architecture of the proposed lightweight Res-CNN is shown in Fig. 2(a).

As illustrated in Fig. 2(a), the input data (I_{in}, V_o) are processed using GhostConv (1-D) augmented with a residual block and a ghost module. The feature map from the GhostConv (1-D) path is concatenated with the feature map from the residual block to enhance model stability. Next, GAP is applied to reduce the dimensionality. The input data (duty ratio, V_o/I_o) are regarded as constants and passed through a dense layer. The output of the dense layer is concatenated with the GAP output and passed through a dense regression head to obtain the final ESR value. This yields the proposed lightweight Res-CNN architecture.

In this model, the ReLU is employed as the nonlinear activation function, as expressed by the following equation:

$$\rho(t) = \max\{0, t\} \tag{3}$$

where $\rho(t)$ is the ReLU activation function. The ReLU, which is used in the hidden layer to introduce nonlinearity, returns the input value if it exceeds zero and returns zero if it is lower than or equal to zero [27].

The architecture of the ghost module used in the lightweight model is illustrated in Fig. 2(b). Many feature maps produced by standard convolution are redundant and can be

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TABLE II
HYPERPARAMETERS OF PROPOSED METHOD

Hyperparameter	Value
Batch size	32
Number of epochs	40
Optimizer	Adam
Initial learning rate	0.001
Dropout rate	0.5
Convolution kernel size	3
Activation function	ReLU
Number of filters in the 1st ghost module	32
Number of filters in the 2nd ghost module	64
Number of filters in the 3rd ghost module	128

approximated effectively using linear transforms. The ghost module operates in three stages. First, standard convolution produces intrinsic feature maps. The ghost module is configured with a ratio of 1.5, which determines the proportion of feature maps generated by the standard convolution and the cheap operation. Next, the intrinsic feature maps are processed through a cheap operation to generate additional feature maps, which can be expressed as follows:

$$y_{i,j} = \Phi_{i,j}(y_i) \tag{4}$$

where y_i' denotes the ith intrinsic feature map and $\Phi_{i,j}$ denotes the cheap operation used to generate the jth ghost feature map. In this study, $\Phi_{i,j}$ is implemented as a depthwise convolution applied independently to each channel without cross-channel mixing. Finally, the feature maps generated via standard convolution and those generated via cheap operation are concatenated to yield the layer output. This procedure preserves similar recognition performance while reducing the computational cost [29]. This design reduces the parameter size, model size, and FLOPs.

Main hyperparameter settings of the proposed model are summarized in Table II, including the batch size, number of training epochs, optimizer, initial learning rate, dropout rate, convolution kernel size, activation function, and the number of filters in the ghost module. The proposed model uses dropout for regularization, with a dropout rate of 0.5 applied after the GAP stage.

A flowchart of the proposed ESR estimation method is shown in Fig. 3. The experimental design incorporates variations in the operating conditions, load conditions, and the DUT to replicate actual operating environments. The evaluation comprises two stages: offline and online. During the offline stage, which is conducted on a PC, the dataset is partitioned into training, validation, and offline test sets. The validation set shares the same DUTs, operating conditions, and load conditions as the training set, whereas the offline test set contains untrained DUTs, operating conditions, and load conditions. Subsequently, data preprocessing with normalization is

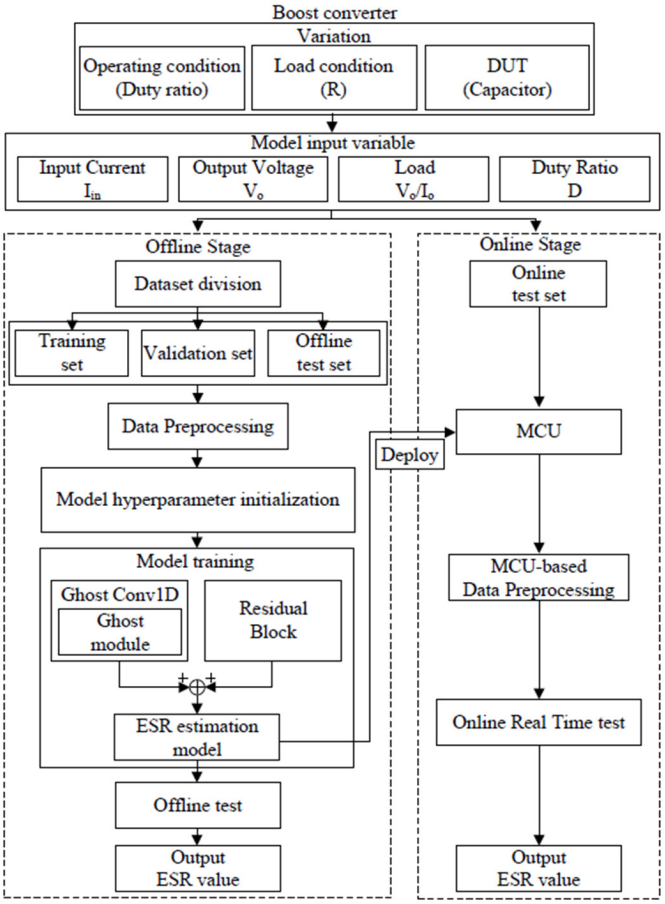


Fig. 3. Flowchart of proposed ESR estimation method.

performed. A mean-centering-based normalization is applied to the input signals by subtracting their mean values.

During the online stage, which is conducted on an MCU, the following tasks are performed. Online real-time ESR estimation is conducted by deploying the lightweight Res-CNN generated in the offline stage to the MCU. Next, the untrained DUTs and operating and load conditions are evaluated to experimentally verify the performance of the model in the online stage.

B. Variations in Input Current and Output Voltage Waveforms

For the same duty ratio and load resistance (R), Fig. 4 illustrates the variations in the input current and output voltage waveforms with capacitor aging. Using Fig. 4(a) as the baseline, Fig. 4(b) shows an ESR increase of 48 %, and Fig. 4(c) shows an increase of 109 %. A higher ESR results in a larger ripple amplitude of the output voltage, as shown in Fig. 4. For a fixed ESR, Fig. 5 shows the input current and output voltage waveforms under different operating and load conditions. Fig. 5(a) shows the waveforms of input current and output voltage for duty ratio = 0.5 and R = 53 Ω, (b) for duty ratio = 0.3 and R = 53 Ω, and (c) for duty ratio = 0.5 and R = 75 Ω.

A comparison between Fig. 5(a) and (b) shows that the duty ratio decreases from 0.5 to 0.3. In this case, the ripple

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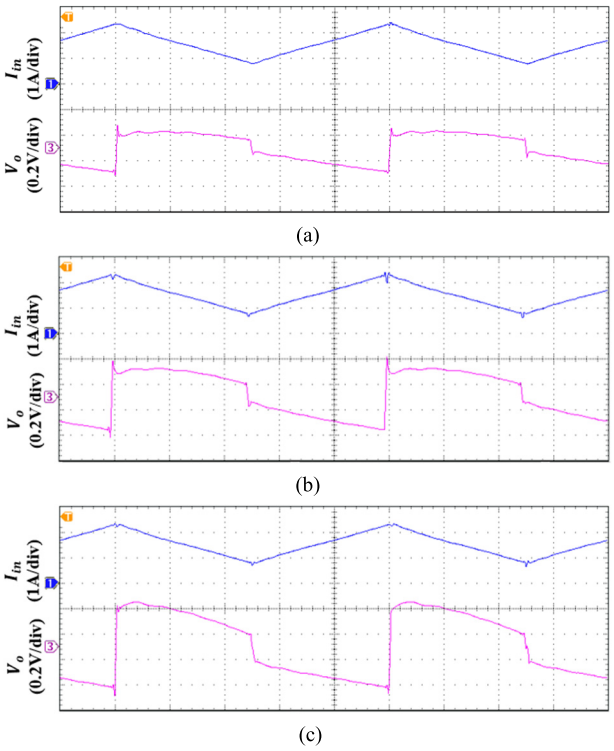


Fig. 4. Experimental waveform of input current and output voltage with (a) ESR = 142.6 mΩ, duty ratio = 0.5, R = 53 Ω. (b) ESR = 211.4 mΩ, duty ratio = 0.5, R = 53 Ω. (c) ESR = 298.7 mΩ, duty ratio = 0.5, R = 53 Ω.

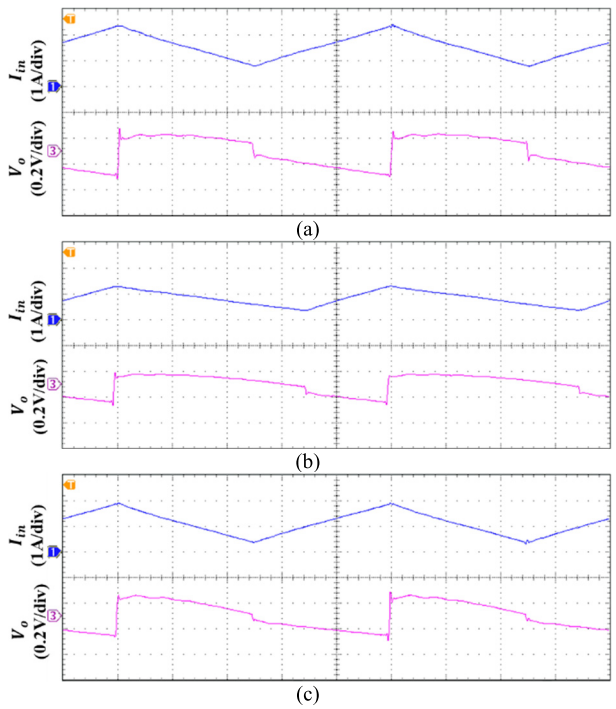


Fig. 5. Experimental waveforms of the input current and output voltage with (a) ESR = 142.6 mΩ, duty ratio = 0.5, R = 53 Ω. (b) ESR = 142.6 mΩ, duty ratio = 0.3, R = 53 Ω. (c) ESR = 142.6 mΩ, duty ratio = 0.5, R = 75 Ω.

amplitudes of the input current and output voltage decrease. A comparison between Fig. 5(a) and (c) indicates that the load resistance increases from 53 to 75 Ω. In this case, the ripple amplitudes of both the input current and the output voltage decrease.

These results indicate that the input current and output voltage waveforms vary not only with capacitor aging but also with the duty ratio and load resistance. Therefore, to accurately diagnose capacitor aging, variations in operating and load conditions should be considered. In this study, a method is developed to diagnose capacitor aging under untrained operating and load conditions.

TABLE III
DUT SETS USED IN THIS STUDY

Number	DUT set	Symbols	Value
1	Training DUT Set	ESR_1, C_1	142.6 mΩ, 112.5 μF
2		ESR_3, C_3	205.4 mΩ, 110.7 μF
3		ESR_5, C_5	247.2 mΩ, 110.5 μF
4		ESR_7, C_7	298.7 mΩ, 109.8 μF
5		ESR_9, C_9	311.7 mΩ, 109.2 μF
6		ESR_{11}, C_{11}	392.1 mΩ, 112.1 μF
7	Test DUT Set	ESR_2, C_2	186.8 mΩ, 113.1 μF
8		ESR_4, C_4	211.4 mΩ, 109.6 μF
9		ESR_6, C_6	277.2 mΩ, 109.8 μF
10		ESR_8, C_8	300.8 mΩ, 110.8 μF
11		ESR_{10}, C_{10}	338.2 mΩ, 110.8 μF

TABLE IV
CONDITION SETS OF THIS STUDY

Number	Condition set	Duty ratio	Load
1	Training Condition Set	0.3	53 Ω
2			75 Ω
3		0.5	53 Ω
4			75 Ω
5	Test Condition Set	0.3	63 Ω
6		0.4	53 Ω
7			63 Ω
8			75 Ω
9		0.5	63 Ω

C. Dataset Composition

The DUTs used for training and testing are listed in Table III. The Training DUT Set is exclusively used for training, whereas the Test DUT Set is exclusively used for testing.

Table IV lists the condition sets. The training condition set comprises conditions used exclusively for training, while the test condition set includes conditions used exclusively for testing.

Table V presents the three evaluation settings (Datasets A, B, and C). The lightweight Res-CNN is trained on the training DUT set and training condition set,

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TABLE V
DATASET CONFIGURATION

Datasets	Train data set	Test data set	Note
Dataset A	Training DUT Set + Training Condition Set	Training DUT Set + Training Condition Set	Train:Validation = 8:2
Dataset B		Test DUT Set + Training Condition Set	Untrained DUT, Trained condition
Dataset C		Test DUT Set + Test Condition Set	Untrained DUT, Untrained condition

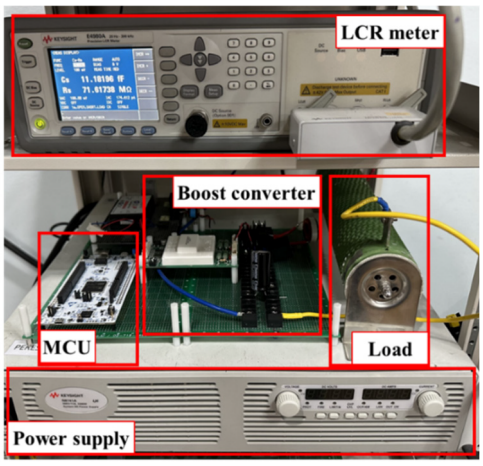


Fig. 6. Hardware experimental platform of the proposed method.

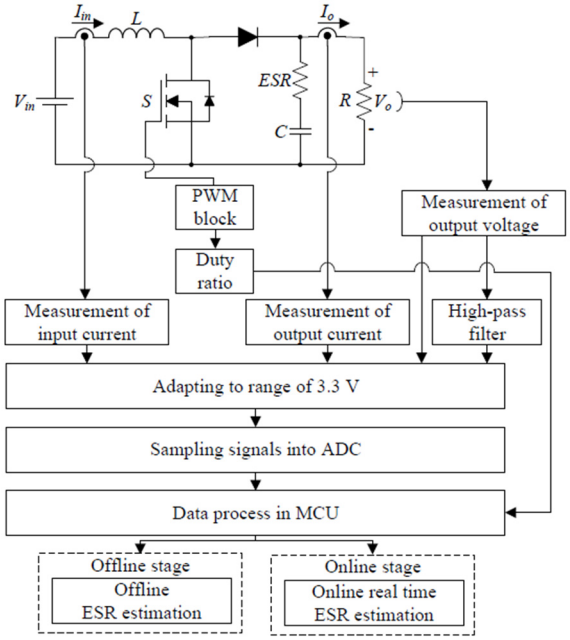


Fig. 7. Proposed lightweight Res-CNN-based ESR estimation system.

while the test sets differ in terms of the generalization scenarios they probe. Dataset A evaluates the trained DUTs under trained conditions, with an 80 %/20 % training-validation split. Dataset B evaluates the untrained DUTs under trained conditions. Dataset C evaluates untrained DUTs under untrained conditions. The difficulty level increases from Datasets A to C; that is, Dataset C is the most challenging dataset. Testing the untrained DUTs and conditions enables an experimental assessment of the generalization performance of the proposed method.

IV. EXPERIMENTS AND RESULT ANALYSIS

A. Experimental Platform Construction

In the boost converter built to train the capacitor aging diagnosis model, the inductor, input voltage, and switching frequency are 220 μH, 20 V, and 25 kHz, respectively. A metal–oxide–semiconductor field-effect transistor (MOSFET) IRFP22N60K is used as the switching device, and diode DHG30I1200HA is employed. The ESR values of the capacitors in the training and the test DUT sets are individually measured using a Keysight E4980A LCR meter with a frequency setting at 25 kHz, which corresponds to the switching frequency of the boost converter. The experimental setup of the proposed online prediction platform is illustrated in Fig. 6.

The lightweight Res-CNN-based ESR estimation system is illustrated in Fig. 7. The input current I_{in} and output current I_o are first level-shifted and scaled to fit within the input range limits (0–3.3 V) of the MCU. The output voltage, V_o, is acquired through two paths. In the first path, the raw signal is conditioned to the MCU range and used to calculate the load resistance, R. In the second path, a high-pass filter extracts the ripple component, following which the signal is conditioned within the MCU range.

The input current, output voltage, and output current are sampled at 200 kHz using a 16-bit ADC configured with an oversampling ratio of 2. Data are acquired from the MCU, and five switching cycles are aggregated to form a single window. Each window comprises the input current, output voltage, output current, and duty ratio collected across the operating, load, and DUT variations. Within each switching cycle, eight samples of the input current, output voltage, and output current are collected. The duty ratio is obtained once per window from the PWM block. We collect 5000 windows per DUT. In this study, experiments are conducted on 11 DUT sets and nine condition sets (see Tables III and IV). Table V summarizes the evaluation datasets: Datasets A, B, and C contain 120 000, 100 000, and 125 000 windows, respectively. The acquired signals are used on a PC in the offline stage to train the lightweight Res-CNN and perform ESR estimation.

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TABLE VI
ACCURACY OF PROPOSED METHOD FOR DIFFERENT DATASETS

Datasets	Condition		Model input variables			
	Duty ratio	Load V_o/I_o	I_{in}, V_o	I_{in}, V_o, Duty ratio	I_{in}, V_o, Load(V_o/I_o)	Proposed
Dataset A (validation)	0.3	53 Ω	51.89%	97.87%	97.42%	98.55%
		75 Ω	51.83%	98.27%	98.31%	97.78%
	0.5	53 Ω	52.00%	98.27%	97.74%	99.24%
		75 Ω	51.47%	97.95%	98.38%	98.07%
	Mean accuracy		51.80%	98.09%	97.96%	98.41%
Dataset B	0.3	53 Ω	50.32%	96.85%	95.96%	98.98%
		75 Ω	51.27%	98.02%	97.60%	96.28%
	0.5	53 Ω	50.08%	95.60%	97.45%	98.06%
		75 Ω	50.87%	97.20%	96.98%	97.53%
	Mean accuracy		50.64%	96.92%	97.00%	97.71%
Dataset C	0.3	63 Ω	49.84%	96.36%	96.09%	97.73%
	0.4	53 Ω	50.75%	89.51%	93.37%	97.54%
		63 Ω	51.19%	91.95%	93.23%	97.60%
		75 Ω	53.67%	95.55%	95.33%	98.05%
	0.5	63 Ω	47.45%	92.94%	96.47%	96.51%
	Mean accuracy		50.58%	93.26%	94.90%	97.49%

In the online stage, the model trained in the offline stage is deployed on the MCU to perform online real-time ESR estimation.

B. Result Analysis

In this section, we present accuracy comparisons based on the model input variables, ablation results with and without residual learning, model performance comparison, and online stage performance evaluation.

1) Accuracy Comparisons by Model Input Variables: Table VI presents the accuracies of Datasets A, B, and C from Table V across the operating and load conditions. Table VI presents the accuracies for each condition across the model input variables for Datasets A (validation), B, and C.

For Dataset A, validation is performed using the training condition set and training DUT set. As presented in Table VI, beginning with two input variables I_{in} and V_o, the accuracy improves across all conditions as additional input variables are included. The proposed method yields the highest per-condition accuracy, with an average of 98.41 %. For Dataset B, testing is performed using the training condition set and test DUT set. Similar to Dataset A, the accuracy increases across all conditions as additional model input variables are included. The proposed method achieves the highest per-condition accuracy, with an average of 97.71 %. For Dataset C, testing is performed using the test condition set and test DUT set. Similar to Datasets A and B, the accuracy increases across all conditions with the inclusion of additional model input variables. The proposed method attains the highest per-condition accuracy, with an average of 97.49 %. From Dataset A to Dataset C, the test setting becomes more challenging, and the mean accuracy decreases correspondingly. Nevertheless, the accuracy of the proposed method on Dataset C remains within one percentage point of that of Dataset A.

Even under the most difficult setting, the proposed method achieves a mean accuracy of 97.49 %, which can demonstrate strong generalization ability under operating conditions, load conditions, and DUT variations. The proposed method is assessed using the R^2 score and RRMSE. The R^2 score and RRMSE are expressed as

$$R^2\text{score} = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \widehat{y}_l\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{5}$$

$$\text{RRMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(\widehat{y}_l - y_i\right)^2}{\sum_{i=1}^{n}\left(\bar{y}_l - y_i\right)^2}} \tag{6}$$

where is the th true value $y_i i, \widehat{y}_i$ denotes the ith predicted value and $\bar{y}$ represents the mean of the true values. The R^2 score curves for the training and validation processes of the proposed method are shown in Fig. 8.

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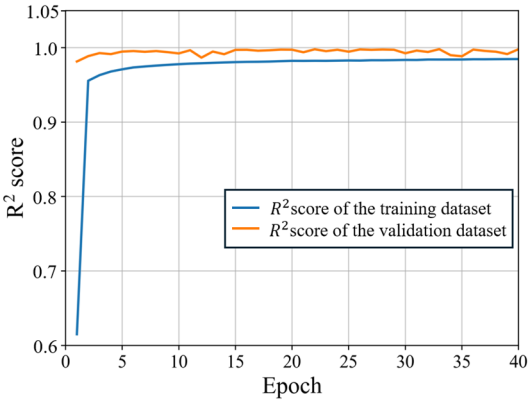


Fig. 8. R^2 score during training and validation processes of proposed method.

TABLE VII
DATASETWISE ACCURACY WITH AND WITHOUT RESIDUAL BLOCK

Datasets	Without residual block	Proposed method (with residual block)
Dataset A	98.41%	98.41%
Dataset B	97.10%	97.71%
Dataset C	94.73%	97.49%

2) Ablation (Effect of Residual Block): Table VII depicts the accuracies of the model without residual block and the proposed method. The "Without residual block" results in Table VII represent the accuracy obtained using the architecture with the residual block removed from the proposed model shown in Fig. 2(a). In this ablation, the only distinction between the two models is the inclusion of a residual block, while the input variables and other settings are kept identical to ensure a fair comparison. It can be shown that incorporating the residual block into the model can improve the accuracy on Datasets B and C. In particular, for Dataset C, which corresponds to the most challenging case, the prediction accuracy increases by approximately 3 %.

3) Analysis of Major Factors Affecting ESR Estimation Performance: To examine the interpretability of the proposed model, group-level Shapley analysis [30] and t-SNE visualization [31] were performed. Shapley analysis is a feature-attribution method based on Shapley values from coalitional game theory that quantifies the contribution of each input feature to the model output. t-SNE is a dimensionality-reduction and visualization technique that embeds high-dimensional feature vectors into a low-dimensional space such that feature vectors that are close in the high-dimensional space tend to remain close in the low-dimensional representation. As shown in Fig. 9, the combined contributions of the V_o and I_{in} waveforms were 80.7 %, 79.4 %, and 85.9 % for Datasets A, B, and C, respectively, confirming that waveform information provides the primary basis for ESR estimation. In Dataset C, the contributions of Load (V_o/I_o) and duty ratio were 8.9 % and 5.2 %, respectively, which is consistent with the larger

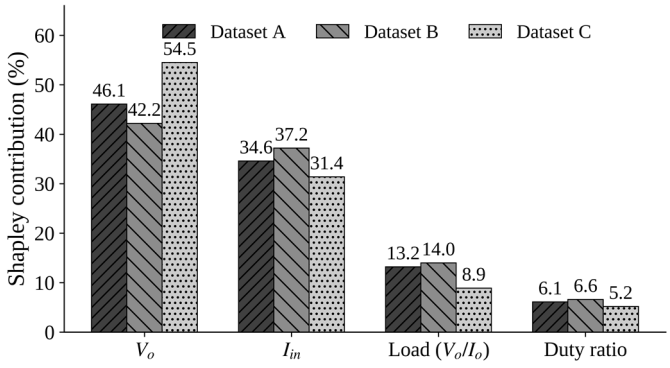


Fig. 9. Shapley contributions of the model inputs to ESR estimation for Datasets A, B, and C.

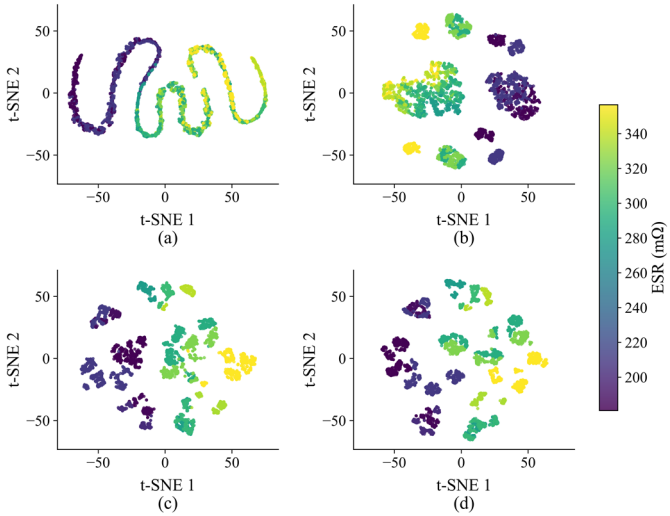


Fig. 10. t-SNE visualization for Dataset C under four input-variable configurations: (a) I_{in} and V_o; (b) I_{in}, V_o, and duty ratio; (c) I_{in}, V_o, and load (V_o/I_o); and (d) I_{in}, V_o, duty ratio, and load (V_o/I_o).

incremental contribution of load than duty ratio in Table VI. Fig. 10 shows that ESR-related feature structure is observed even when only I_{in} and V_o are used, and the feature separation by ESR level becomes clearer when condition information is included. These results indicate that the proposed model primarily uses ESR-sensitive waveform information while employing operating/load-condition information as auxiliary input information.

4) Verification and Comparison of Performance of Proposed Method: Table VIII presents comparisons between the proposed method and seven comparison AI models based on three metrics: mean accuracy, R^2 score, and RRMSE. All the comparison models are evaluated using the same input variables (I_{in}, V_o, duty ratio, and V_o/I_o) as in the proposed method to ensure fairness. The proposed method shows high performance in terms of the mean accuracy values of 98.41 % and 97.49 % on Datasets A and C, respectively. Dataset A mirrors the training setup with the same DUTs and conditions, whereas Dataset C is the most challenging generalization set containing both untrained DUTs and untrained conditions. The decrease from Dataset A to Dataset C is within one percentage point, indicating strong generalization. On Dataset C, the proposed

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TABLE VIII
PERFORMANCE COMPARISONS OF PROPOSED AND PREVIOUS METHODS

Model	Dataset A			Dataset B			Dataset C		
	Mean accuracy (%)	R^2 score	RRMSE	Mean accuracy (%)	R^2 score	RRMSE	Mean accuracy (%)	R^2 score	RRMSE
KNN	99.97	1.000	0.001	95.39	0.924	0.276	90.56	0.815	0.416
ResNet-50	98.49	0.996	0.062	96.27	0.958	0.192	94.67	0.933	0.239
LSTM	98.90	0.998	0.042	97.41	0.976	0.152	96.11	0.958	0.186
GPR	69.92	0.005	0.998	77.99	-0.004	1.002	75.04	0.002	0.999
1D CNN+LSTM	98.54	0.998	0.044	96.29	0.950	0.219	94.52	0.936	0.245
SVM	45.30	-1.662	1.629	54.94	-2.961	1.990	58.70	-1.918	1.690
MobileNet	99.07	0.999	0.029	97.87	0.986	0.113	97.54	0.985	0.117
Proposed method	**98.41**	**0.998**	**0.048**	**97.71**	**0.984**	**0.119**	**97.49**	**0.983**	**0.126**

TABLE IX
COMPLEXITY COMPARISON OF PROPOSED AND CONVENTIONAL METHODS

Model	Parameter size (k)	Model size (MB)	FLOPs (M)
KNN	-	35.48	27.03
ResNet-50	4,139.27	15.87	55.08
LSTM	76.43	0.571	6.08
GPR	-	96.74	2.86
1D CNN+LSTM	127.43	0.63	10.13
SVM	107.96	71.27	53.55
MobileNet	101.22	0.4	7.62
Proposed method	**21.87**	**0.099**	**1.72**

TABLE X
MEAN ACCURACY OF PROPOSED METHOD UNDER DIFFERENT NOISE CONDITIONS

Datasets	Without noise	With 5% noise	With 10% noise
Dataset A	98.41%	96.44%	93.35%
Dataset B	97.71%	96.23%	94.52%
Dataset C	97.49%	97.31%	94.72%

method also achieves high performance with R^2 score and RRMSE values (0.983 and 0.126, respectively).

Furthermore, comparison of the generalization performance shows that the differences among the models become more apparent as the evaluation difficulty increases from Dataset A to Dataset C. On Dataset C, MobileNet and the proposed method achieve similar mean accuracies of 97.54 % and 97.49 %, respectively; however, the proposed method exhibits the smallest decrease in mean accuracy from Dataset A to Dataset C (0.92 percentage points), compared with 1.53 percentage points for MobileNet, indicating stable generalization under untrained DUT and operating/load conditions.

Table IX presents comparisons of model complexity using three indicators: parameter size, model size, and FLOPs. ResNet-50 yields the largest parameter size, approximately 189 times that of the proposed method. The proposed method has at least 71 % fewer parameters than the other models. Because KNN and GPR are

Nonparametric, a fixed finite-parameter vector does not describe their complexity precisely; therefore, parameter size alone is insufficient. KNN, GPR, and SVM models exhibit larger model sizes than the other models, whereas the proposed method achieves more than a 75 % reduction in model size relative to the other models. In addition, the FLOPs values are lower than those of the comparison models. Overall, the proposed lightweight Res-CNN yields the lowest values across all three complexity indicators-parameter size, model size, and FLOPs, yielding an embedded-friendly tradeoff between computational complexity and predictive performance. In particular, it can be seen that the developed model shows better performance than MobileNet, which is known as a lightweight neural network architecture [32]. Compared with MobileNet, the proposed method reduces the parameter count, model size, and FLOPs by factors of approximately 4.6, 4.0, and 4.4, respectively. Despite the substantial reduction in model complexity, the proposed method achieves nearly the same estimation accuracy as MobileNet, as shown in Table VIII. Thus, the proposed method demonstrates competitive estimation performance with lower model complexity than the lightweight model.

Experiments are conducted to show the robustness of the proposed method against noise. Fig. 11 depicts the experimental waveforms of the input current and the output voltage in the boost converter, with no noise, 5 % noise, and 10 % noise added. Those waveforms with different noise levels are used for the ESR estimation, and the prediction results are shown in Table X. From Table X, as the noise level increases to 10 %, the proposed method still shows around 95 % diagnostic

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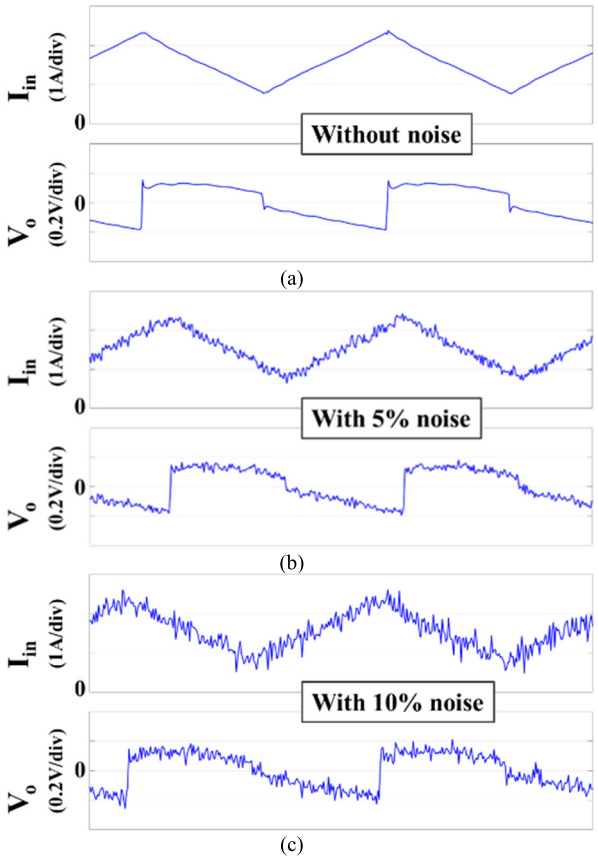


Fig. 11. Experimental waveforms of input current and output voltage with different noise conditions: (a) without noise, (b) with 5 % noise, and (c) with 10 % noise.

accuracy, although the accuracy is slightly reduced compared with that under no noise. Thus, the developed model preserves a high level of diagnostic performance even with the input signals corrupted.

The effects of input-voltage fluctuation on ESR estimation accuracy are tested to evaluate the generalization capability of the proposed model. The input voltage variation ratio is defined as the ratio of the actual input voltage to the nominal input voltage, as expressed by

$$\textit{Input voltage variation ratio}\,(\%) = \frac{V_{in,actual}}{V_{in,nominal}} \times 100. \quad (7)$$

The input voltage is varied within ±20 % of the nominal input voltage. Fig. 12 shows the mean accuracy of the proposed model under the input voltage variation. As shown in Fig. 12, the proposed method shows only about a 3 % reduction in mean accuracy even under a 20 % reduction in the input voltage.

To evaluate the robustness of the proposed model against switching-frequency variation, tests were conducted on Dataset C at 22.5, 25, 27.5, and 30 kHz. The mean accuracies at these switching frequencies were 98.45 %, 96.80 %, and 95.23 %, and 93.19 %, respectively, with an overall mean accuracy of 95.92 %. Thus, the proposed method maintained a high accuracy above 93 % over the tested switching-frequency range.

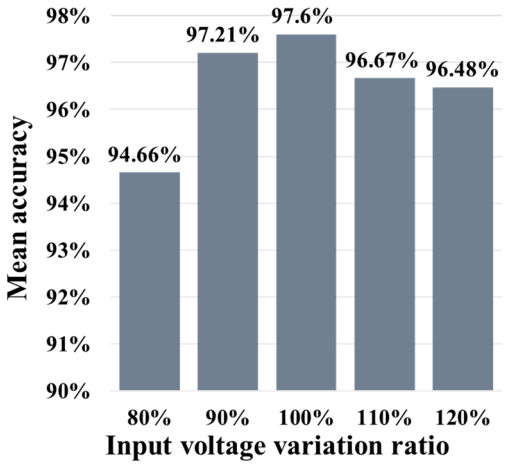


Fig. 12. Mean accuracy of proposed method under input voltage variation for Dataset C, duty ratio = 0.4 and R = 63 Ω.

TABLE XI
ONLINE REAL-TIME ACCURACY RESULTS OF PROPOSED MODEL

Datasets	Condition		Accuracy
	Duty ratio	Load	
Dataset B	0.3	53 Ω	97.43%
		75 Ω	96.74%
	0.5	53 Ω	97.16%
		75 Ω	98.91%
	Mean accuracy		97.56%
Dataset C	0.3	63 Ω	96.08%
	0.4	53 Ω	98.64%
		63 Ω	97.66%
		75 Ω	98.77%
	0.5	63 Ω	95.64%
	Mean accuracy		97.36%

Table XI presents the online real-time ESR estimation results by condition for Datasets B and C, as defined in Table V, using the proposed method. The online stage yields mean accuracy values of 97.56 % for Dataset B and 97.36 % for Dataset C (see Table XI). The offline and online cases exhibit a similar trend, with Dataset B achieving a higher mean accuracy than Dataset C. The online real-time setting runs on the MCU during converter operation without a train or validation split. Consequently, Dataset A is excluded from the online real-time test, and results are reported only for Datasets B and C. In the online stage, the inference time is 10.908 ms. This represents a decrease of more than 15 % relative to the version that does not use the ghost module.

The mean accuracies of the proposed method in the offline and online stages are presented in Fig. 13. The online real-time evaluation attains a mean accuracy of 97.56 % for Dataset B and 97.36 % for Dataset C in Fig. 13. The differences relative

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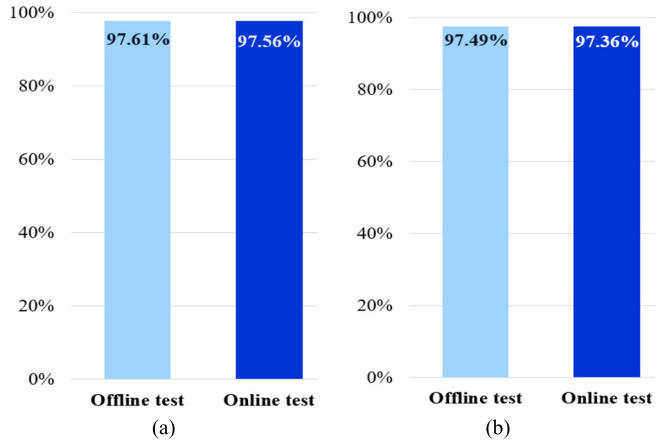


Fig. 13. Accuracy comparison of proposed method at offline and online stages for (a) Dataset B and (b) Dataset C.

to the PC-based offline results are at most 0.2 percentage points, confirming the close agreement between the online and offline performances.

Overall, the proposed method offers the advantages of strong generalization performance, low model complexity, and feasibility for real-time MCU-based implementation. The proposed method maintains high ESR estimation performance under untrained DUT and operating/load conditions while using a lightweight model structure. In addition, the model was successfully implemented on the STM32H753ZI MCU for online real-time ESR estimation, confirming the feasibility of MCU-based deployment. However, the current model is trained using labeled data collected in a laboratory environment, which may limit its direct applicability to practical field environments where sufficient labeled data are difficult to obtain. Future work will investigate adapting the laboratory-trained model to a target field domain through transfer learning using limited field-labeled data to improve its practical field applicability.

V. Conclusion

This article presents a lightweight Res-CNN regression network that targets simultaneous generalization and lightweight design for capacitor condition estimation in a DC/DC boost converter. The generalization is evaluated for various DUTs, operating conditions, and load conditions by expanding the set of input variables and incorporating residual learning. In the most demanding setting with an untrained DUT and untrained conditions, the proposed method attains a mean accuracy of 97.49 %, demonstrating strong robustness. Among the seven comparison models, the proposed method achieves the highest generalization performance, with the decrease in mean accuracy from validation to the untrained setting kept below one percentage point. In addition, applying the ghost module for model lightweighting yields significant model reductions in terms of the parameter size (by at least 71 %), model size (by at least 75 %), and FLOPs (by at least 39 %), resulting in the lowest overall complexity. The model is also deployed on an MCU in an online stage for real-time evaluation. The inference time is reduced by more than 15 % relative to the version without the ghost module, and the mean accuracy for the untrained DUT and operating/load conditions is 97.36 %, in close agreement with the offline results. Overall, the proposed method satisfies the requirements for generalization, lightweight design, and real-time operation, making it suitable for MCU-based field deployment. Therefore, the proposed lightweight Res-CNN is suitable for online diagnostic platforms of ECs in DC/DC boost converters.

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